

Inverse Folding Models Sample Sequence Space Shaped by MHC Anchor Identity and TCR Context

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Abstract

Predicting which peptides a given MHC and T-cell receptor pair will accommodate is a major challenge. Inverse-folding models recover native interface residues well above chance, but a single average recovery rate obscures which structural constraints actually drive that signal. Here we applied four inverse-folding models to twenty solved pMHC–TCR structures under two contexts, generating 1.6 million peptide designs scored against two independent binding predictors. Recovery is bimodal: P2, which requires a specific side chain, recovers well above average, while PΩ, a permissive backbone constraint, recovers well below it despite being equally load-bearing; ProteinMPNN-family models sample stronger predicted HLA-A*02:01 binders than ESM-IF1. Removing the T-cell receptor increases diversity and predicted affinity but lowers recovery at experimentally validated TCR-contact positions and shifts each model’s internal confidence. Together, these results establish inverse-folding sampling distributions, not just point-estimate recovery, as a general probe of the constraints governing peptide recognition by MHC and TCR.

1 Introduction

Determining which peptide sequences a given major histocompatibility complex (MHC) and T-cell receptor (TCR) pair can jointly accommodate is a central problem in T-cell immunology, with direct consequences for vaccine design, cancer immunotherapy, and TCR-engineered cell therapies. A solved pMHC–TCR crystal structure fixes one specific peptide out of an astronomically large sequence space, and recovering which sequences a structure-based model would propose in its place is not straightforward, because peptide recognition depends jointly on MHC-groove side-chain specificity and TCR surface complementarity, and a given position along the peptide can be constrained by one, the other, or both, depending on structural context. Most peptides presented by a given MHC allele and read by a given TCR remain uncharacterized, and distinguishing which positions are truly load-bearing for MHC binding versus TCR recognition, rather than simply asserting that "anchor residues matter," remains difficult from structure alone.

Experimental approaches to this problem, such as deep mutational scanning of peptide libraries against a fixed

pMHC–TCR pair, can measure binding and functional consequences directly, but each such screen is specific to the one complex under study and does not scale to the combinatorics of peptide, MHC allele, and TCR identity. Structure-based deep learning offers a complementary, more scalable approach, since inverse-folding models, trained to recover the sequence most compatible with a fixed backbone, implicitly learn the packing and geometric constraints that make a given structure stable, without ever training on a binding-affinity label. On the receptor side of this same interface, this approach has recent precedent. Comparing ProteinMPNN and ESM-IF1 against physics-based Rosetta design across a panel of 38 solved TCR:pMHC complexes, Ribeiro-Filho et al. (2024) found that both inverse-folding models recover native CDR3 interface residues well above chance, establishing that inverse folding captures real, non-trivial signal from a TCR–pMHC interface.

That result addresses only one side of the interface, whether the receptor can be redesigned around a fixed peptide and MHC. Whether the same class of models can recover the peptide itself, and what their full sampling distribution over peptide sequences reveals about the constraints governing this interface, is the complementary question we address here. Because inverse-folding models score compatibility with structure rather than a labeled binding phenotype, evaluating both a model’s point-estimate recovery and its broader sampling distribution serves two purposes at once. It is a direct test of the model’s capability as a sequence-design tool, and, because MHC-groove geometry and TCR-contact geometry are structurally distinguishable positions along the same peptide, that same output can serve as a structural probe of which positions a model treats as side-chain-constrained versus permissive.

To address this question with sufficient power, we assembled a panel of twenty independent pMHC–TCR structures, four inverse-folding tools, two structural contexts (TCR present or removed), and two independent sources of validation, namely computational binding-affinity predictors and experimental TCR-binding measurements.

2 Methods

2.1 Panel and models

Twenty pMHC–TCR crystal structures were retrieved from the Protein Data Bank (Supplementary Table S2 lists all twenty accession codes), each with complete inverse-folding runs under both structural contexts and all four tools. Every structure includes a bound peptide, the MHC class I heavy chain, β_2 -microglobulin, and a complete TCR $\alpha\beta$ heterodimer resolved in the same asymmetric unit. Fifteen present a 9-mer peptide, four present a 10-mer, and one (4MJI) presents an 8-mer. Nineteen structures are HLA-A*02:01; the remaining structure, 4MJI, is HLA-B*51:01 and was excluded from every analysis that depends on an allele-specific binding predictor, since scoring a B*51:01-restricted peptide against an A*02:01 model would be uninformative rather than merely noisy.

Each of the four tools redesigns only the peptide chain, with the MHC, β_2 -microglobulin, and (where present) TCR chains held fixed as structural context. Two context conditions were run for every structure, **full** (MHC + β_2 m + TCR $\alpha\beta$) and **mhonly** (the TCR removed from the input structure entirely, not merely masked). Each condition was sampled to 10,000 raw designs at a fixed sampling temperature of $T = 0.1$.

2.2 Metrics and statistics

Recovery is the fraction of raw designs matching the native residue at a position. **Diversity** is unique-peptide fraction and per-position Shannon entropy. **Predicted binding affinity** uses MHCflurry and ESMCBA, neither exposed to this panel during training, on every unique design from the nineteen HLA-A*02:01 structures. **External TCR-binding validation** draws on SKEMPI 2.0 and IEDB, matched to each structure’s crystallized TCR by CDR3.

The four models are **ProteinMPNN**, a message-passing GNN encoding the fixed backbone as a k -nearest-neighbor graph and decoding sequence autoregressively; **ProteinMPNN (no MHC)**, the same architecture retrained with MHC/TCR-class structures excluded; **ESM-IF1**, a GVP-GNN encoder plus transformer decoder trained substantially on AlphaFold2-predicted backbones; and **LigandMPNN**, a ProteinMPNN variant with explicit ligand/heteroatom context nodes (Supplementary Table S1).

3 Results

3.1 Models Sample Strong Position-Specific Sequence Bias

To establish how much of the peptide sequence space these models actually explore, we sampled 10,000 designs per (structure, model) cell in the pMHC+TCR context and counted how many were distinct. Only 50 designs per cell were unique on average (0.50% of the 10,000 drawn, ranging from 1 to 240 across the 80 cells), so these models return redundancy rather than diversity (Supplementary Figure S2). Sequence logos show every model imposes strong position-specific restriction independent of recovery (Supplementary Figures S3–S4d), a property of the sampling distribution rather than evolutionary conservation, tracing the

same contacts seen in the crystals, where 23 of 40 MHC positions contacting the peptide anywhere in the panel do so in all twenty (Supplementary Figure S1c).

3.2 Recovery is bimodal, and the two MHC anchors do not behave the same way

To determine which peptide positions carry recoverable signal, we measured per-position recovery of each crystal’s own native residue, pooled across twenty structures and four models. Recovery is bimodal rather than uniform, and the two positions where this matters most are the two positions immunology already identifies as anchors (Figure 1). P2, seated in pocket B, constrains a specific side chain and recovers well above the interior average (0.64 vs. 0.49), whereas P Ω , seated in pocket F, imposes a permissive backbone constraint and recovers significantly *below* it (0.19 vs. 0.49, $p = 0.0054$). Both are structurally indispensable, yet only P2 gives a design model a signal it can exploit.

To ask whether that asymmetry reflects side-chain identity alone or a coarser physicochemical preference, we compared each native anchor residue against its own peptide’s interior positions on the Kyte-Doolittle hydrophobicity scale, and separately scored every design for whether it preserved the native residue’s side-chain class (Supplementary Figure S12). Both anchors are markedly more hydrophobic than the interior they sit alongside (3.44 at P2 and 3.65 at P Ω , against an interior mean of 0.79; $p = 0.0013$ and $p = 0.0001$, paired Wilcoxon across 19 structures), confirming that both pockets recruit hydrophobic residues. The two anchors nonetheless part company on how tightly that preference is enforced in the designs. P2 designs preserve the native side-chain class well above the interior baseline (85.6% vs. 70.7%, $p = 0.019$), whereas P Ω designs do not (62.4% vs. 70.7%, not significant), even though P Ω designs do show significantly reduced hydrophobicity variance relative to the interior ($p = 0.041$). P Ω therefore behaves as a pocket that biases toward a hydrophobic character without fixing which hydrophobic residue satisfies it, which is what a low exact-identity recovery at a structurally load-bearing position should look like. We note that a discrete side-chain classification counts cysteine as polar despite its hydrophobic Kyte-Doolittle value, and four of the nineteen native peptides carry cysteine at P Ω , so the continuous scale is the more reliable of the two readouts here.

Averaged across the entire peptide, the four models fall within a narrow band of mean recovery (0.41 to 0.47 in full context and 0.28 to 0.36 without the TCR; Supplementary Figure S5), narrow enough that the models might appear indistinguishable by inspection. A Friedman test indicates otherwise in both contexts ($\chi^2 = 10.80$, $p = 0.013$, full context; $\chi^2 = 12.18$, $p = 0.007$, mhonly), so this apparent similarity reflects proximity in magnitude rather than statistical equivalence. As an aside, on 3QDG the central Gly-Ile-Gly motif at positions 4–6, well characterized in the literature, recovers essentially perfectly here (1.00, 0.73, 1.00 vs. 0.49), though we have not attempted a systematic motif annotation across the full panel.

The relationship between resolution and recovery illustrates how pooling across a heterogeneous panel can un-

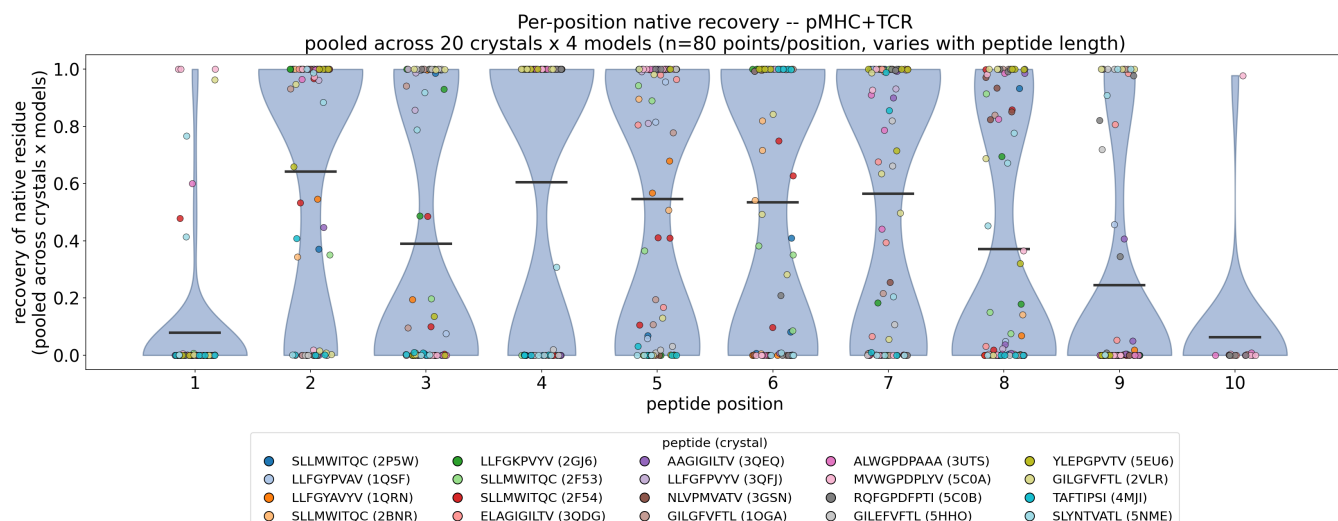


Figure 1: **Per-position recovery is bimodal, and the two MHC anchors diverge.** Recovery of the native residue, pooled across all 20 crystals and 4 models, one violin per peptide position (pMHC+TCR context). Individual points are colored by crystal. P2 sits well above the interior average; P Ω (the rightmost populated position for each peptide length) sits well below it; this is the opposite of what "both are MHC anchors" would predict if anchoring alone determined recoverability.

derstate a genuine effect. Pooled across all twenty distinct peptides, the correlation is modest, and only one model survives correction individually ($r = -0.49$ pooled, $p = 0.030$; ESM-IF1 alone, $r = -0.63$, $p = 0.003$; Supplementary Figure S6). However, four of the twenty structures (2P5W, 2BNR, 2F53, and 2F54) are independent crystals of the identical peptide (SLLMWITQC, the NY-ESO-1 epitope, a cancer-testis antigen widely targeted in adoptive T-cell and TCR-engineered immunotherapies) bound to nearly the same TCR clone, differing primarily in resolution (1.9–2.7 Å). Holding peptide identity fixed in this way removes a confound that the pooled panel cannot, and the effect becomes unambiguous. Declining resolution drives recovery down for two of the four models ($r = -0.99$, $p = 0.012$, ProteinMPNN (no MHC); $r = -0.97$, $p = 0.031$, ESM-IF1) and drives unique design counts up for the remaining two ($r = 0.97$, $p = 0.028$, ProteinMPNN; $r = 0.96$, $p = 0.040$, ProteinMPNN (no MHC); Supplementary Figure S7). The pooled-panel estimate was therefore not incorrect, but diluted by heterogeneity.

3.3 ProteinMPNN-Family Models Enrich for Predicted Strong pMHC Binders

Having established the recovery rates of the native peptide and the sequence space each structure samples, we next asked whether those sampled peptides would actually be presented, scoring every unique design from the nineteen HLA-A*02:01 structures for predicted HLA-A2 binding with two unrelated predictors, MHCflurry and ESM-CBA, neither of which had seen this panel (Figure 2).

The two predictors agree closely on the same designs ($r = 0.83$ – 0.91 per model in full context, up to 0.94 without the TCR), and agreement between two methodologically unrelated predictors constitutes stronger evidence

than either predictor alone. On this shared landscape, projected jointly into one UMAP embedding space (Supplementary Figure S11), ProteinMPNN and LigandMPNN consistently sample toward higher predicted affinity than ESM-IF1. At the 500 nM cutoff, 61–78% of ProteinMPNN-family and LigandMPNN designs qualify by MHCflurry, compared with 45% for ESM-IF1 ($\chi^2 = 238.46$, $p = 2 \times 10^{-51}$; see also Section 3.4). This pattern is better characterized as ESM-IF1 being a consistent low-affinity outlier than as a general superiority of ProteinMPNN over ESM-IF1, since the other three tools do not share this deficit.

3.4 Removing the TCR Increases Diversity and Predicted Affinity, but Lowers Recovery

To understand the effect of the TCR on what these models propose, we resampled every structure with the TCR deleted from the input and compared the resulting designs against the full-complex designs above.

Every model explores substantially more sequence space without the TCR, with unique design counts roughly doubling or tripling (ProteinMPNN 1081 $\rightarrow$ 3671; ProteinMPNN (no MHC) 958 $\rightarrow$ 2362; ESM-IF1 1069 $\rightarrow$ 1865; LigandMPNN 668 $\rightarrow$ 2111; Supplementary Figure S5). This larger space skews toward higher predicted affinity, but not uniformly across models. ProteinMPNN (no MHC) shows the largest shift (strong-binder rate 77.5% $\rightarrow$ 95.2%), LigandMPNN a smaller but genuine one (61.1% $\rightarrow$ 70.7%), while ProteinMPNN and ESM-IF1 show little change (Supplementary Figure S10). The affinity gain associated with TCR removal thus appears concentrated in the two models already most context-sensitive, rather than reflecting a general consequence of removing the TCR.

A related but distinct signal comes from each model's own confidence in the sequences it designs, independent

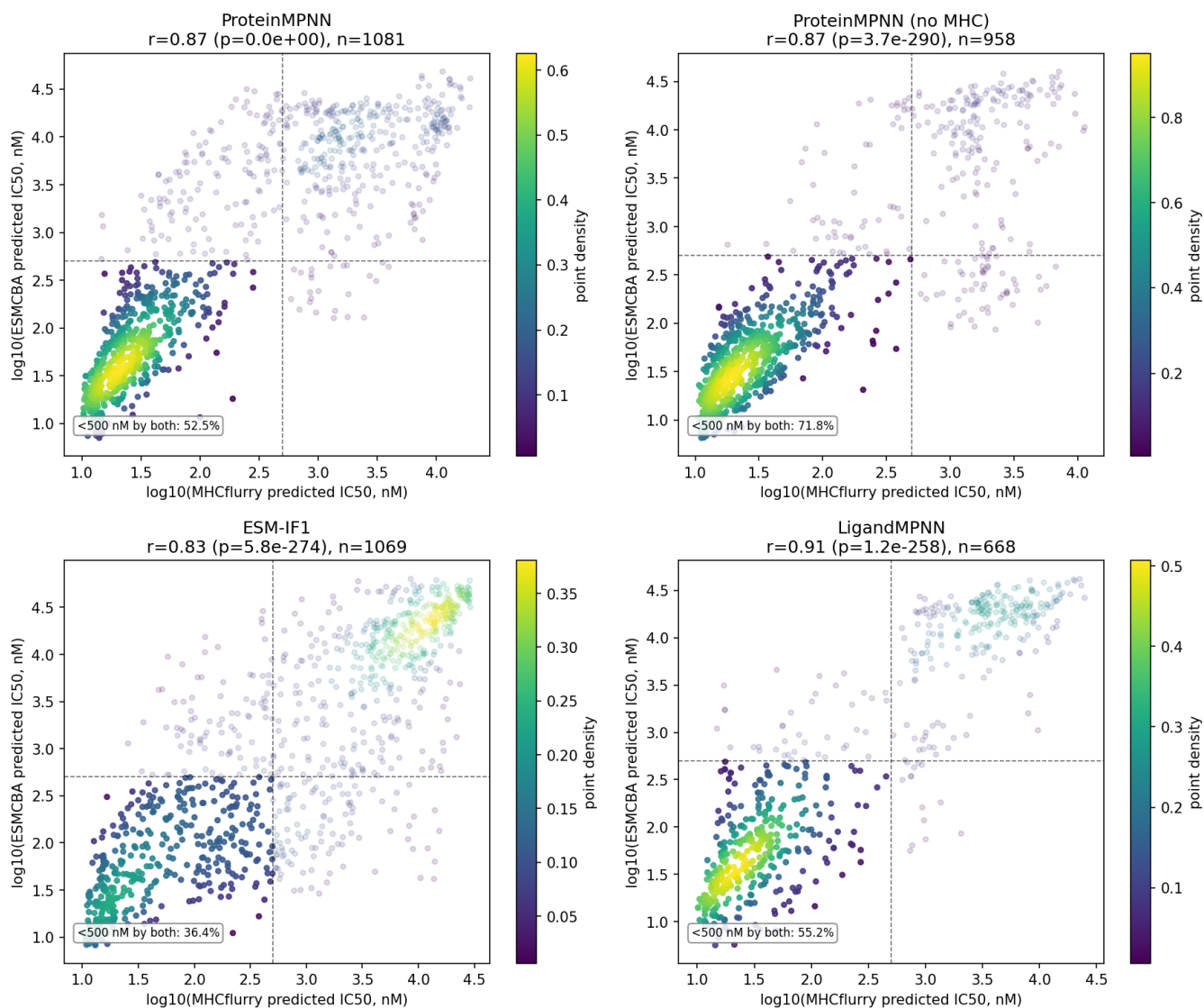


Figure 2: **MHCflurry and ESMCBA agree closely on which designs bind tightly.** $\log_{10}(\text{predicted IC}_{50}, \text{nM})$ for both predictors, pMHC+TCR context, one panel per model. Dashed lines mark the 500 nM strong-binder cutoff on both axes; only designs strong by *both* predictors jointly (bottom-left quadrant) are rendered opaque, everything outside that quadrant is faded. The percentage is the joint strong-binder rate, meaning the fraction of designs under 500 nM by both predictors at once.

of either external predictor. ProteinMPNN and ProteinMPNN (no MHC) report an average per-residue negative log-likelihood for each design, for which lower values indicate higher confidence; LigandMPNN reports an overall confidence score, for which higher values indicate higher confidence. Pooled across all nineteen HLA-A*02:01 structures, removing the TCR shifts this internal score for every model, and in every case toward lower self-assessed confidence (ProteinMPNN mean $1.016 \rightarrow 1.295$; ProteinMPNN (no MHC) $0.925 \rightarrow 1.128$; LigandMPNN $0.413 \rightarrow 0.321$; Mann-Whitney $p < 10^{-200}$ for all three; Supplementary Figure S9a). This shift is not uniform across the panel. Broken out per structure and Bonferroni-corrected, it reaches significance in 15–17 of 19 structures depending on the

model, but not in the remaining two to four (Supplementary Figure S9b). That a model can become less confident in its own designs by its own internal metric while simultaneously being scored as a stronger binder by an external predictor that never saw those designs indicates that the two signals are not redundant, since one reflects how comfortably a sequence fits the model's learned distribution, the other how likely it is to bind, and removing the TCR does not move them the same way. This distinction is more informative than amino-acid-level recovery or composition alone, since it speaks to what the model itself, not just an external predictor, "believes" about the sequences it is generating.

3.5 Recovery Loss Concentrates at Experimentally Validated TCR-Contact Positions

To understand how these residue interactions contribute functionally to TCR binding, we tested whether the recovery lost without the TCR concentrates at positions that experimental data already implicate in receptor engagement. We cross-referenced SKEMPI 2.0 and IEDB against each structure's crystallized TCR clone (matched by CDR3 sequence) and identified 21 (structure, position) pairs with an experimentally measured mutation that alters TCR binding. Pooling recovery across exactly these positions, TCR-present designs recover the native residue 54–58% of the time, compared with 34–38% for TCR-absent designs (Figure 3), a consistent, though not universal, gap; a minority of individual positions move in the opposite direction. The clearest single illustration lies outside this panel, at position 4 of the MEL5/ELAGIGILTV complex, where SKEMPI confirms that alanine substitution abolishes TCR binding outright, recovery in our data collapses from 98% with the TCR to 0% without it.

4 Discussion

The unifying observation of this study is that the claim "MHC anchor positions matter" is true yet, on its own, of limited explanatory value. P2 and P Ω are both structurally indispensable, yet behave oppositely under inverse folding because one is a side-chain readout and the other is predominantly a backbone constraint tolerant of many side chains. This distinction, rather than a blanket statement about anchor positions, is what a sequence-design model can meaningfully report. The same caution regarding pooled analyses applies to resolution, where a weak, borderline correlation pooled across twenty distinct peptides conceals an unambiguous, model-specific effect once peptide identity is controlled for using a same-peptide replicate set, the four independent SLLMWITQC/NY-ESO-1 crystals. The TCR-removal results likewise resist a single summary verdict by construction, since diversity increases for every model, affinity increases primarily for two of them, and recovery decreases on average but not at every position, because "does the model require the TCR" was never a single, unified question.

5 Limitations and scope

- **$T=0.1$ only.**
- **Allele skew.** Nineteen of twenty structures are HLA-A*02:01; the one exception (4MJI, HLA-B*51:01) is also the hardest structure in the panel by recovery, and with $n = 1$ we cannot separate allele identity from TCR identity or structure-specific difficulty.
- **Predicted affinity is not measured affinity.** Agreement between MHCflurry and ESMCBA constitutes real evidence, but neither substitutes for an experimental binding assay.
- **Not all TCR-binding evidence decouples MHC loading from TCR engagement equally well.** Only SPR-

based dissociation-constant measurements are performed on an already-folded pMHC complex; functional assays (cytotoxicity, activation) cannot distinguish a peptide that ceased binding MHC from one that ceased being recognized by the TCR.

- **Recovery is not validated binding.** It measures agreement with a crystallized peptide, not experimentally confirmed activity of the designed sequences.
- **The MEL5/ELAGIGILTV case study is outside the 20-structure panel** and uses a separate, earlier design campaign; it's a sharper single-structure illustration, not panel-scale evidence.

6 Data and code availability

All data and code are available in the if-mhc GitHub repository: <https://github.com/sermare/if-mhc>.

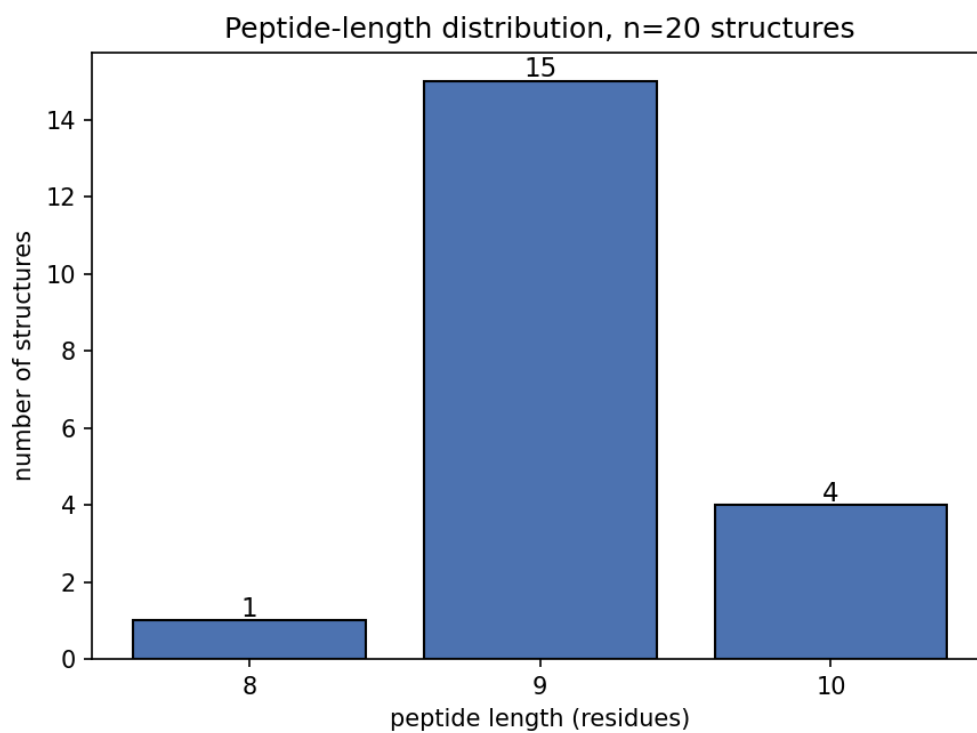


Figure S1a: **Peptide length distribution** across all 20 structures in the panel.

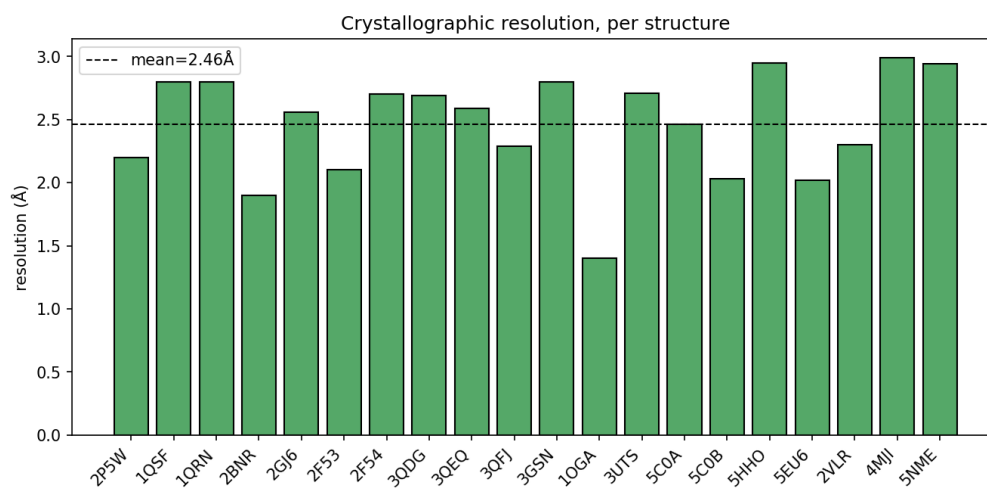


Figure S1b: **Crystallographic resolution per structure.**

Model	Params	Architecture	Training data	Predicted structures
ProteinMPNN	1.66M	Message-passing GNN	PDB (pdb_2021aug02), general backbones	–
ProteinMPNN (no MHC)	1.66M	Message-passing GNN (identical to above)	Same corpus, MHC/TCR-class structures excluded	–
ESM-IF1	141.7M	GVP-GNN + transformer	CATH-derived backbones	+12M structures AF2
LigandMPNN	2.62M	Message-passing GNN + ligand context	PDB, augmented with explicit ligand and/heteroatom context	–

Table S1: **Models compared.** Parameter counts verified directly from each checkpoint’s `state_dict`. ESM-IF1 is the only tool whose training set is built substantially from AlphaFold2-predicted structures rather than solved crystal structures (~12M predicted backbones, on top of its CATH-derived crystal set).

PDB	Peptide	Length	Allele	Resolution (Å)
2P5W	SLLMWITQC	9	HLA-A*02:01	2.20
1QSF	LLFGYPVAV	9	HLA-A*02:01	2.80
1QRN	LLFGYAVYV	9	HLA-A*02:01	2.80
2BNR	SLLMWITQC	9	HLA-A*02:01	1.90
2GJ6	LLFGKPVYV	9	HLA-A*02:01	2.56
2F53	SLLMWITQC	9	HLA-A*02:01	2.10
2F54	SLLMWITQC	9	HLA-A*02:01	2.70
3QDG	ELAGIGILTV	10	HLA-A*02:01	2.69
3QEQ	AAGIGILTV	9	HLA-A*02:01	2.59
3QFJ	LLFGFPVYV	9	HLA-A*02:01	2.29
3GSN	NLVPMVATV	9	HLA-A*02:01	2.80
1OGA	GILGFVFTL	9	HLA-A*02:01	1.40
3UTS	ALWGPDPAAA	10	HLA-A*02:01	2.71
5C0A	MVWGPDPLYV	10	HLA-A*02:01	2.46
5C0B	RQFGPDFPTI	10	HLA-A*02:01	2.03
5HHO	GILEFVFTL	9	HLA-A*02:01	2.95
5EU6	YLEPGPVTV	9	HLA-A*02:01	2.02
2VLR	GILGFVFTL	9	HLA-A*02:01	2.30
4MJI	TAFTIPSI	8	HLA-B*51:01	2.99
5NME	SLYNTVATL	9	HLA-A*02:01	2.94

Table S2: **The twenty-structure panel.** PDB accession, native peptide, peptide length, MHC allele, and crystallographic resolution for every structure in the panel.

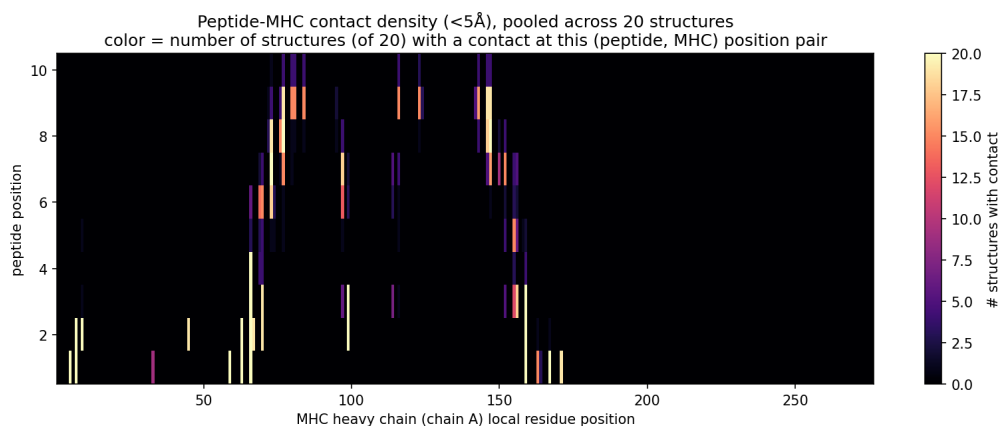


Figure S1c: **MHC residue vs. peptide contact density**, pooled across all 20 structures.

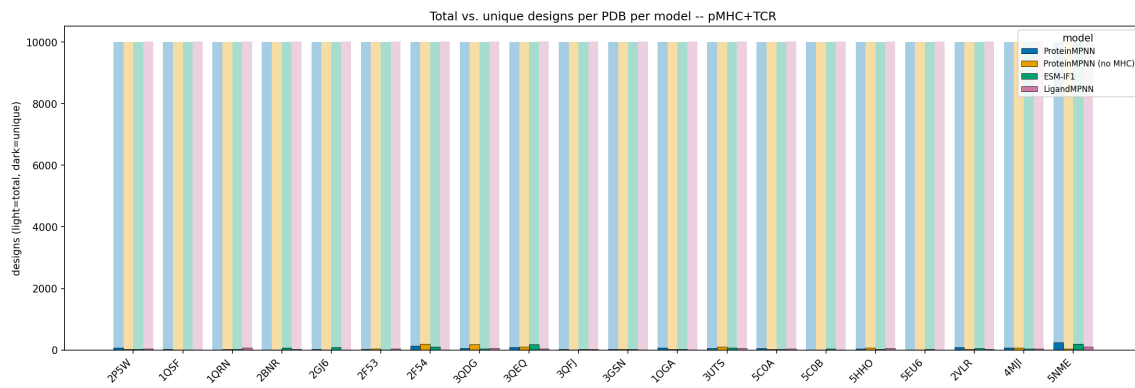


Figure S2: **Extreme redundancy at $T=0.1$.** 10,000 raw draws per (structure, model); the unique fraction is 0.50% on average and never exceeds a few percent.

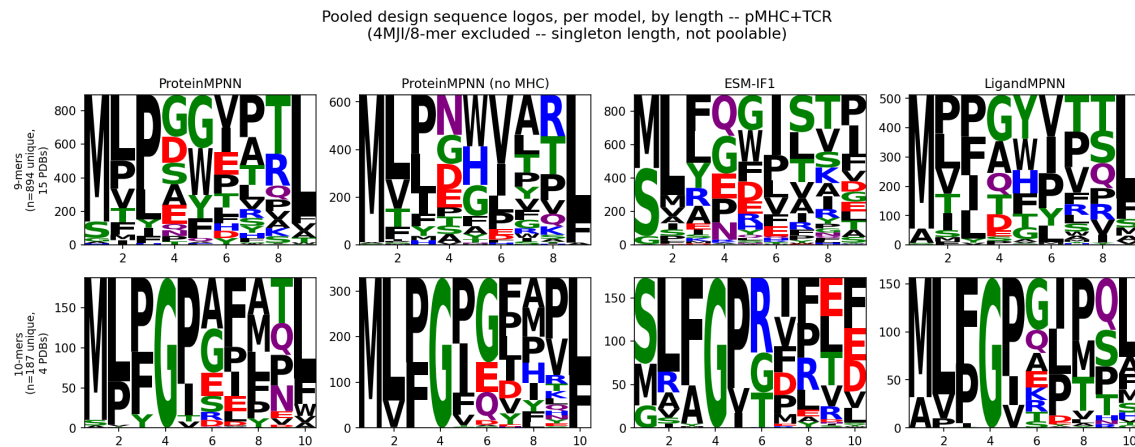


Figure S3: Design sequence logos, pooled by peptide length, one column per model.

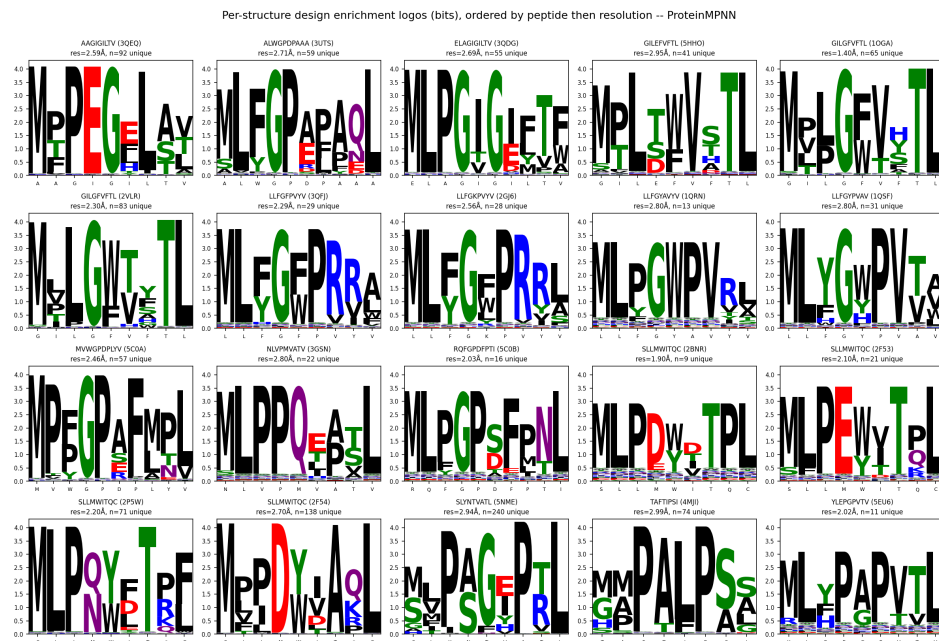
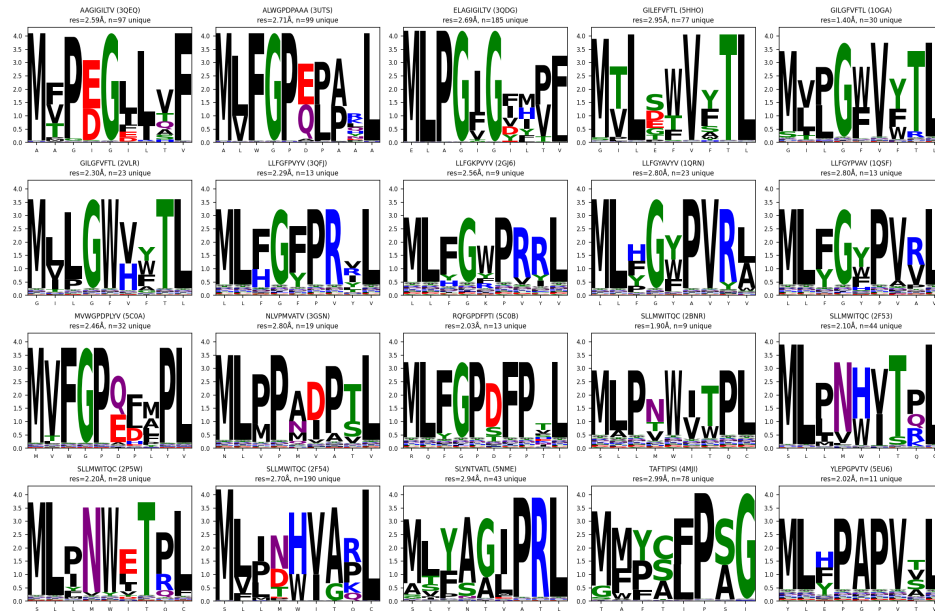
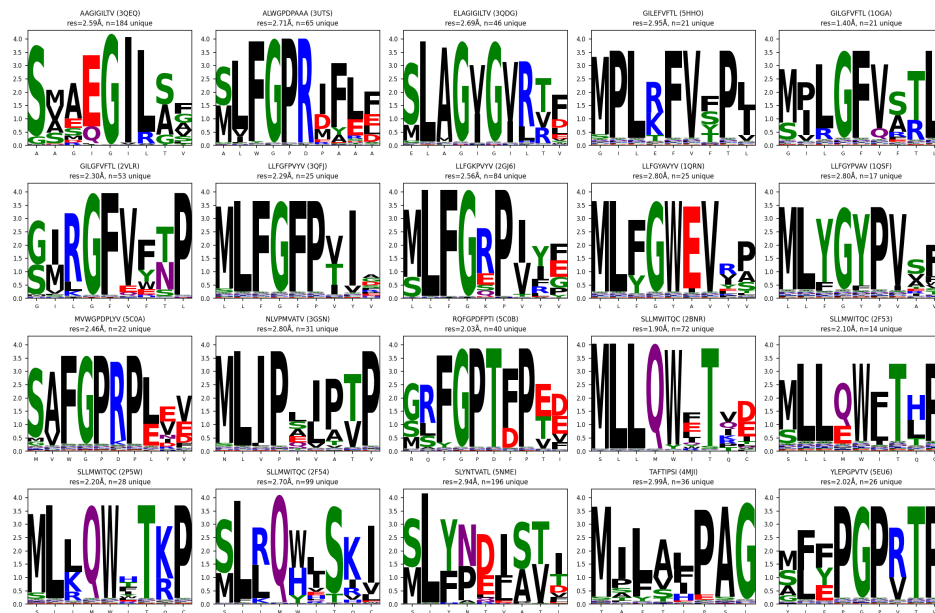


Figure S4a: **Per-structure design logos, ProteinMPNN.** All 20 structures, ordered by native peptide then resolution.

Per-structure design enrichment logos (bits), ordered by peptide then resolution -- ProteinMPNN (no MHC)



Per-structure design enrichment logos (bits), ordered by peptide then resolution -- ESM-IF1



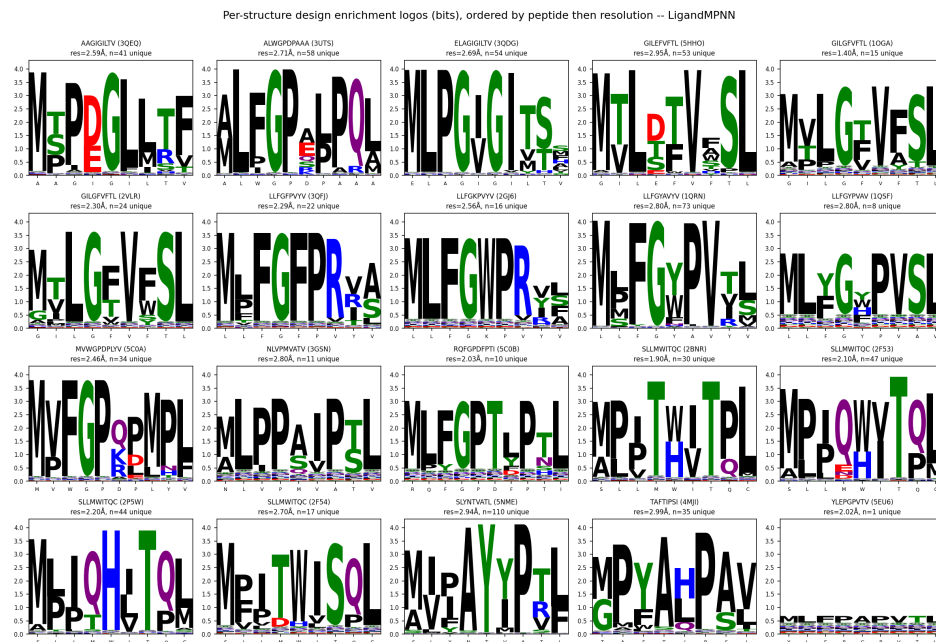


Figure S4d: Per-structure design logos, LigandMPNN.

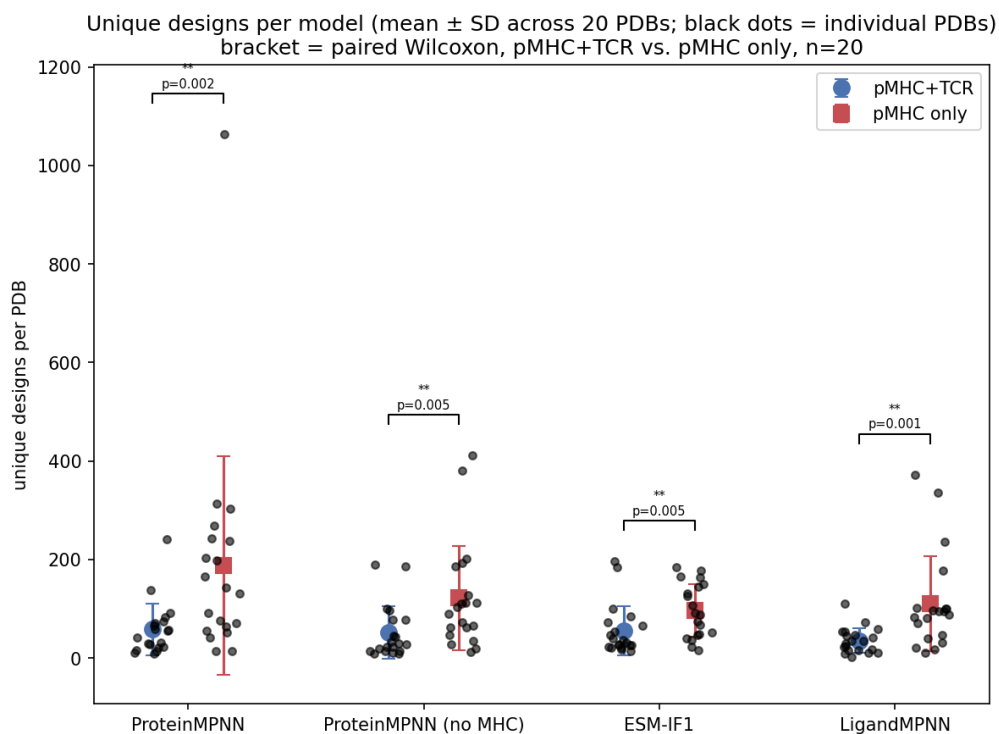


Figure S5: Unique designs per model, mean $\pm$ SD across structures, pMHC+TCR vs. pMHC only.

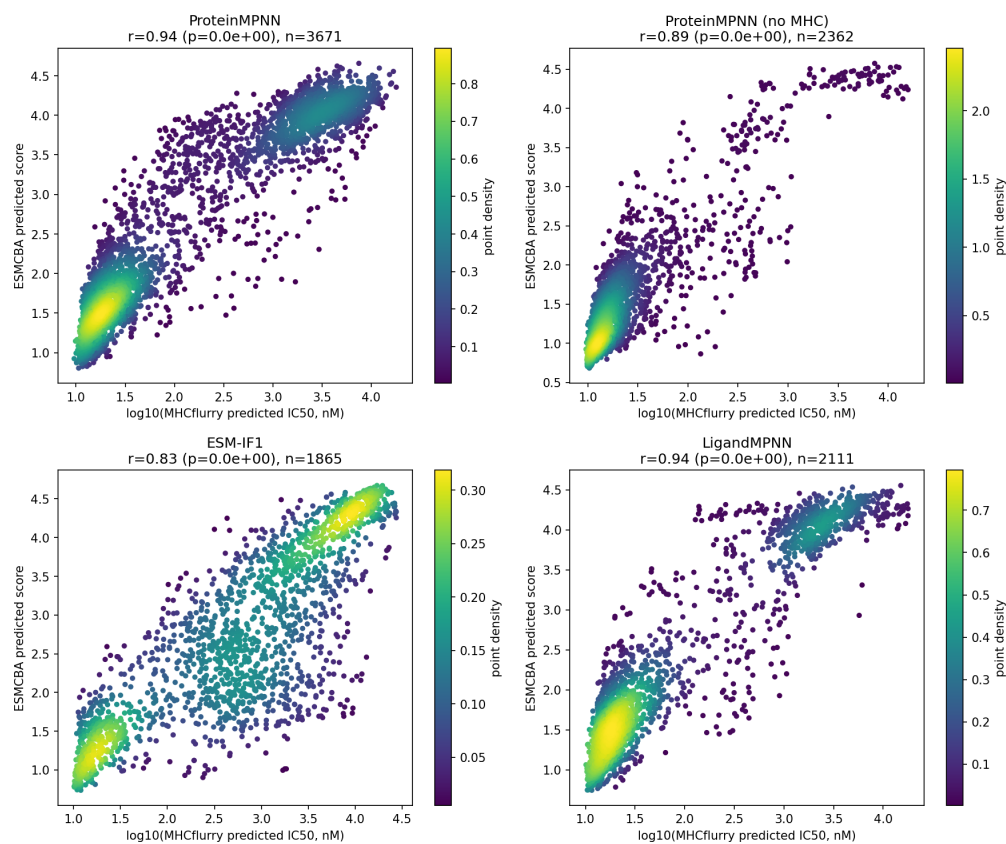


Figure S8a: ESMCBA vs. MHCflurry agreement, pMHC-only designs, by model.

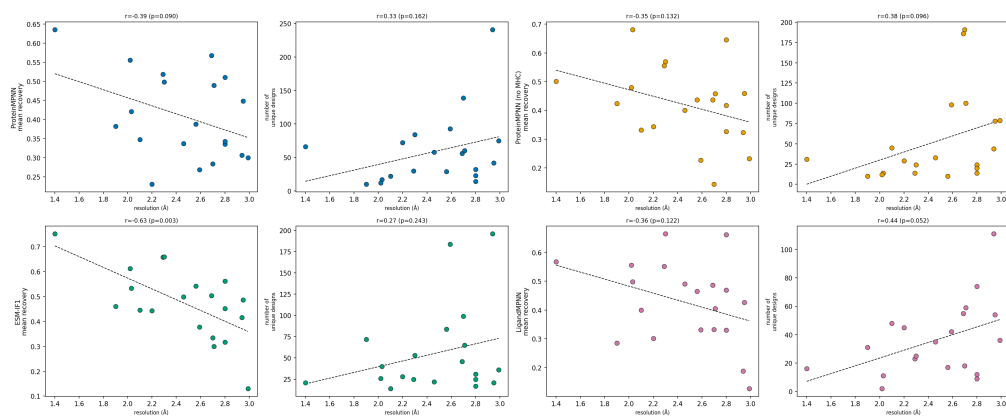


Figure S8b: Per-model resolution/design-count relationship, full panel.

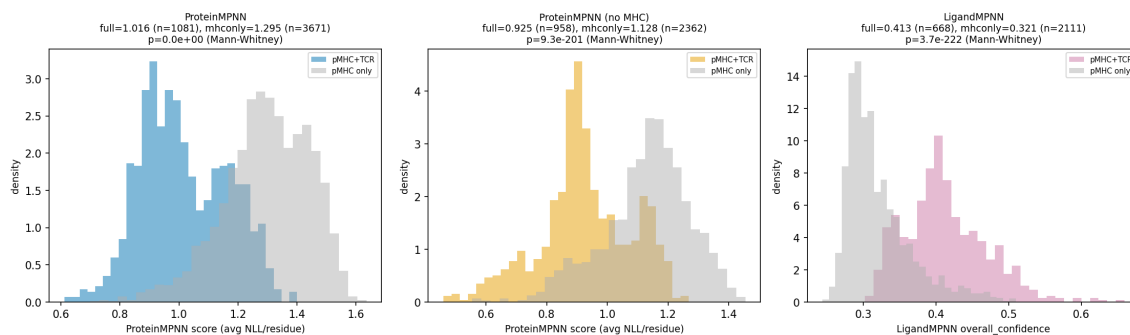


Figure S9a: **Each model's own confidence score for its designs shifts when the TCR is removed**, pooled across all nineteen HLA-A*02:01 structures. ProteinMPNN and ProteinMPNN (no MHC): average per-residue negative log-likelihood, lower is more confident. LigandMPNN: overall confidence, higher is more confident. All three shift toward lower self-assessed confidence without the TCR (Mann-Whitney $p < 10^{-200}$ in all three cases).

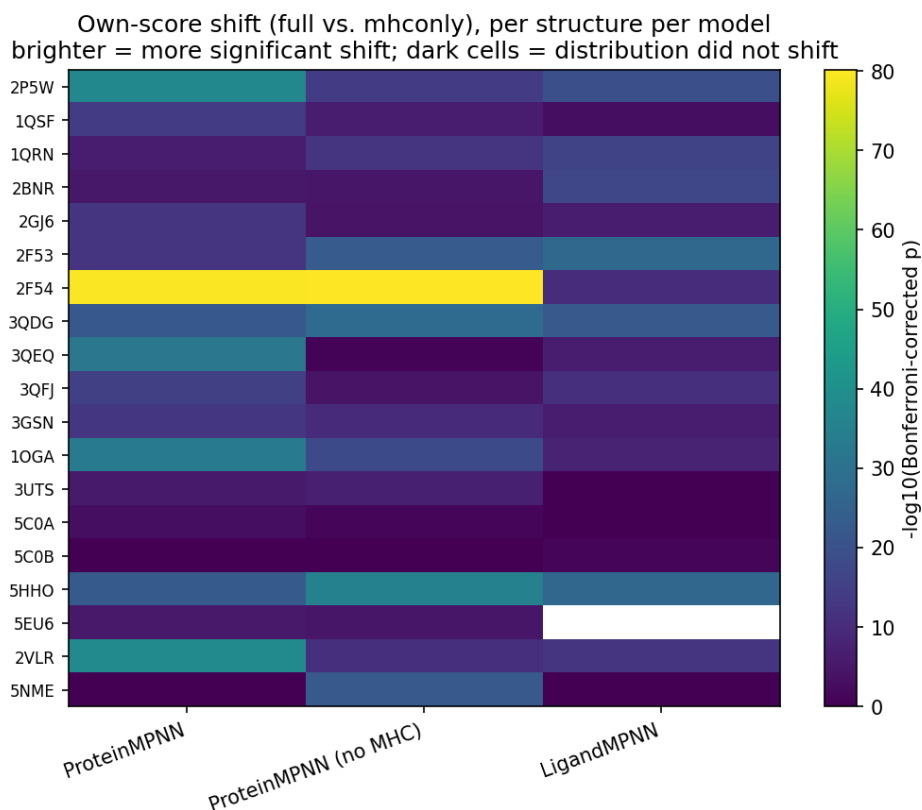


Figure S9b: **The own-score shift is not uniform across the panel.** Bonferroni-corrected significance (full vs. mhonly) per structure per model; dark cells are structures where the model's own-score distribution did not shift.

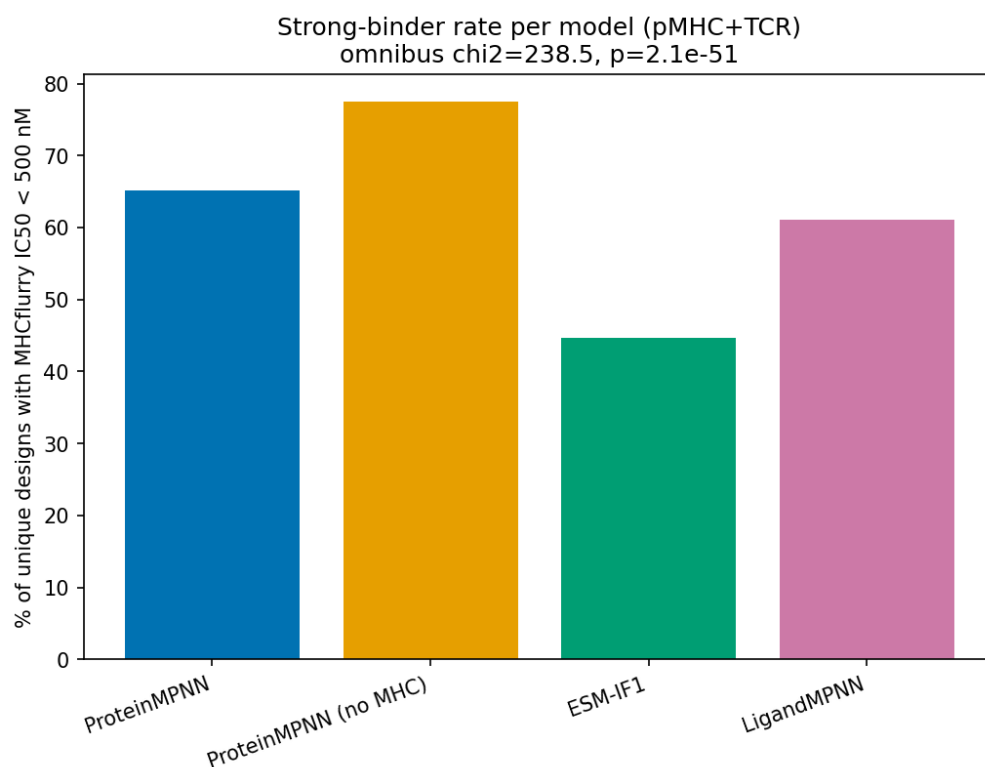


Figure S10: **Removing the TCR raises predicted affinity, mostly for two of the four models.** Strong-binder rate (MHCflurry IC50 < 500 nM), full vs. pMHC-only context, Wilson 95% CI. ProteinMPNN (no MHC) and LigandMPNN both shift up substantially without the TCR; ProteinMPNN and ESM-IF1 barely move. ESM-IF1 is the lowest-affinity model in both conditions.

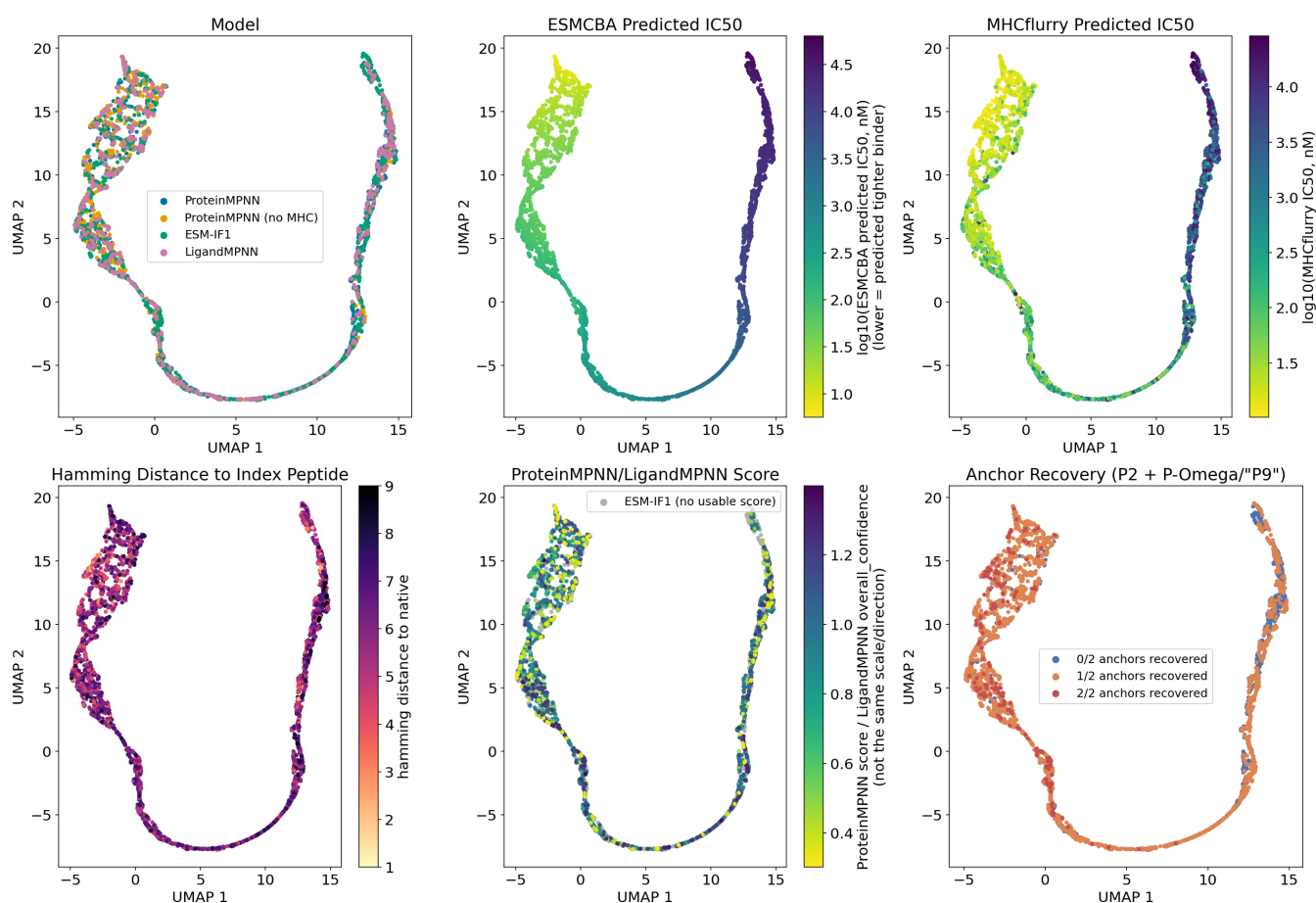


Figure S11: **The sampled sequence space, colored by predicted binding affinity.** One UMAP (ESMCBA embeddings, fit jointly over every unique design), recolored six ways. All four models occupy the same overlapping landscape rather than four separate clusters, and MHCflurry and ESMCBA, built differently and trained differently, trace the identical gradient across it.

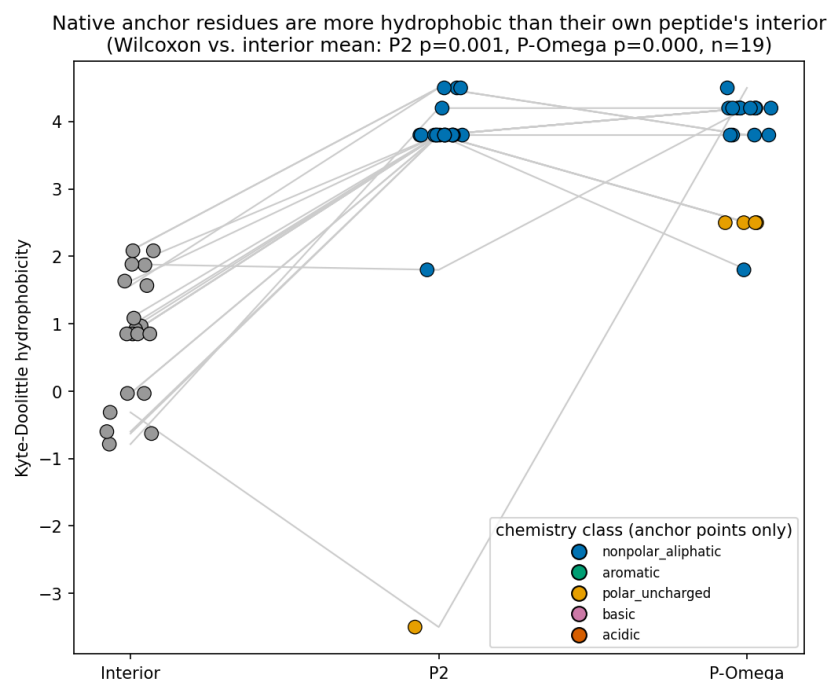


Figure S12: **Both MHC anchors are more hydrophobic than the peptide interior they sit alongside.** Kyte-Doolittle hydrophobicity of the native residue at P2 and P Ω for each of the 19 HLA-A*02:01 structures, plotted against that same peptide's mean over its own interior positions (P3 through the second-to-last, excluding P1). Grey lines connect the three values belonging to one peptide; anchor points are colored by side-chain class. Both anchors sit well above their own interior (paired Wilcoxon, $p = 0.0013$ for P2 and $p = 0.0001$ for P Ω , $n = 19$), so the two pockets do not differ in whether they prefer hydrophobic residues. They differ in how strictly the designs honor that preference, which is the distinction reported in Section 3.2. The four points at P Ω colored as polar are the cysteine-terminated SLLMWITQC/NY-ESO-1 replicates, a residue whose discrete classification understates its hydrophobicity.